

Fig. 2.5. A white dwarf (*left*), neutron star (*middle*), and black hole (*right*) that all weigh as much as 1.2 Suns. For the white dwarf I show only a tiny segment of its surface.

Since stars are generally no heavier than about 100 Suns, the black holes to which they give birth are also no heavier than 100 Suns. The giant black holes in the cores of galaxies, a million to 20 billion times heavier than the Sun, therefore, cannot have been born in the death of a star. They must have formed in some other way, perhaps by the agglomeration of many smaller black holes; perhaps by the collapse of massive clouds of gas.